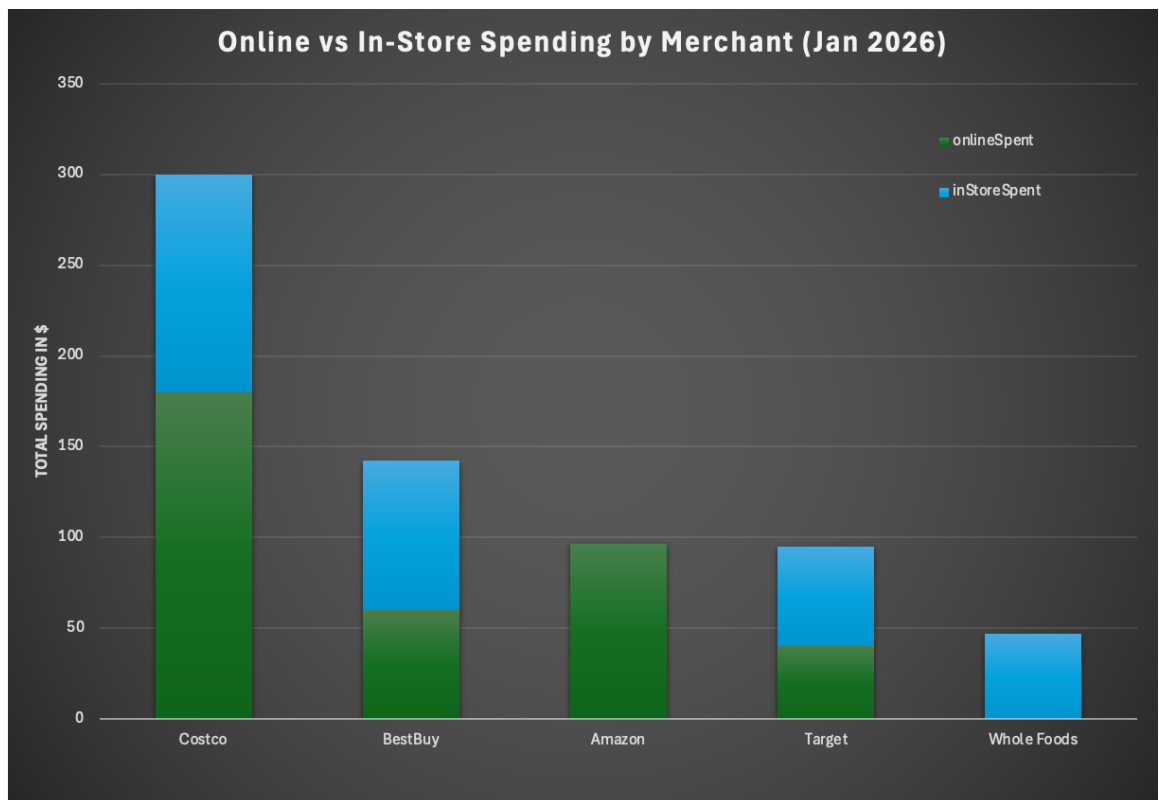


Data Visualizations

Create two visualizations of the data in your database. Clearly explain the data story conveyed by each visualization. Ensure that the visualizations and data stories are useful and appropriate given the intended use of your database. Add the SQL to your SQL script.

Visualization 1:



Data Story: Online vs In-Store Spending by Merchant (January 2026 – Selected User):

This chart presents the January 2026 spending patterns for the selected user in the Smart Budget Database System, highlighting total expenditure by merchant and channel preference (online vs. in-store).

Costco accounts for the highest total spending, with approximately \$300 in combined purchases. A significant portion of Costco spending occurred in-store, suggesting bulk or routine household purchases made physically rather than online.

BestBuy ranks second in total spending, with a more balanced split between online and in-store purchases. This pattern may indicate electronics or higher-value items where the user compares options online before completing purchases.

Amazon spending appears entirely online, reinforcing its role as a purely digital marketplace in this dataset. This suggests convenience-driven purchases or recurring online household orders.

Target shows moderate spending distributed across both channels, indicating flexible purchasing behavior depending on product type or urgency.

Whole Foods reflects the lowest overall spending and is exclusively in-store, which aligns with typical grocery shopping behavior where fresh items are selected in person.

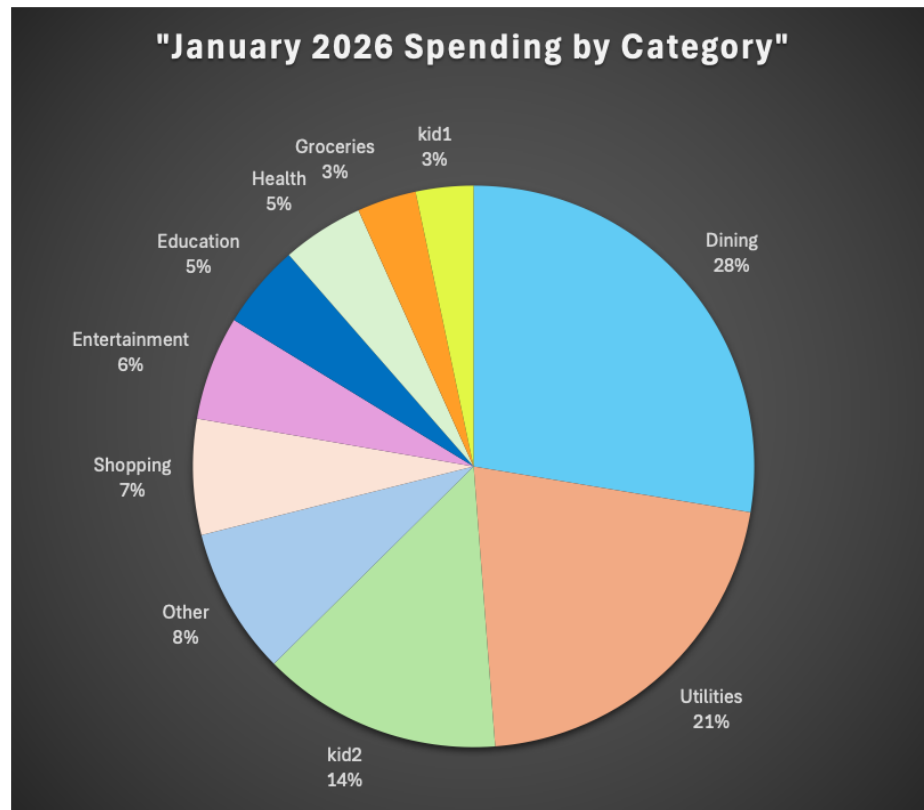
Overall Insight (Selected User Perspective):

For the selected user:

- Warehouse and grocery purchases (Costco, Whole Foods) skew heavily toward in-store transactions.
- Technology and general retail merchants (BestBuy, Target) show hybrid channel behavior.
- Amazon remain fully online.

This distribution highlights how channel preference for the selected user is influenced by product category, convenience, and purchasing habits within the household.

Visualization 2:



Data Story: Spending Distribution by Category (January 2026 – Selected User):

This pie chart illustrates how total spending for January 2026 for the selected user is distributed across various spending categories. The visualization highlights which categories contribute most significantly to the selected user's monthly budget.

The largest portions of spending are concentrated in Dining (28%) and Utilities (21%), indicating that recurring household costs and food-related expenses account for the majority of the selected user's expenditures during the month.

A noticeable share of spending is also allocated to kid2 (14%) and Other (8%), suggesting that household-specific and miscellaneous purchases play a meaningful role in the selected user's overall financial activity. These categories represent flexible spending areas that may fluctuate month to month depending on family needs.

Moderate spending appears in Shopping (7%) and Entertainment (6%), reflecting discretionary spending patterns that complement essential expenses. Meanwhile, smaller proportions are observed in Health (5%), Education (5%), Groceries (3%), and kid1 (3%), indicating controlled and targeted spending in these areas for the selected user.

Overall Insight (Selected User Perspective):

For the selected user, essential categories such as **Dining and Utilities** have the greatest impact on total monthly spending. Because these categories represent the largest shares, even small adjustments in these areas could significantly influence overall budget performance.

At the same time, discretionary categories remain proportionally smaller, indicating a relatively balanced spending approach. The distribution reflects a household budget that prioritizes necessities while maintaining controlled spending in lifestyle and optional categories.

Monitoring high-impact categories regularly can support more effective budget management and long-term financial stability for the selected user.